

Predict Everything: From RGB Sequences to 3D Objects

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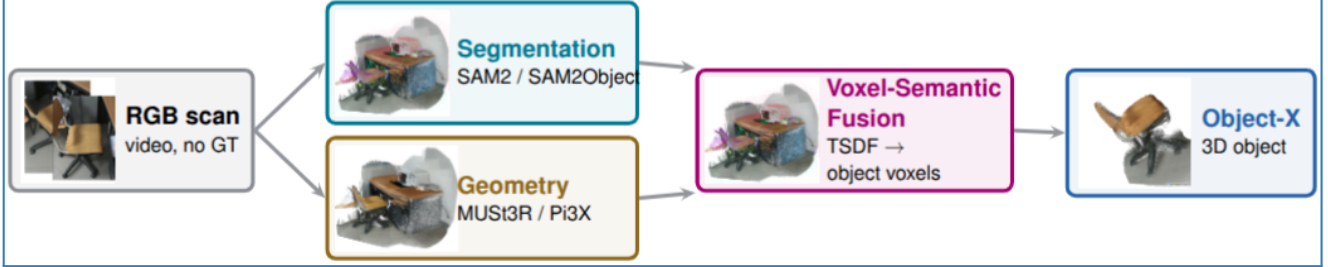


Figure 1. **Pipeline Overview.** Our weakly-supervised framework prepares multi-view sequences for 3D reconstruction without requiring ground truth information. Given an RGB-only sequence, the framework dynamically predicts multi-view segmentation masks and geometric properties directly from 2D frames. These predictions are then fused into consistent 3D object voxels, providing a robust initialization that can be directly leveraged by downstream reconstruction architectures like Object-X.

Work distribution

Team member	Share	Main contribution
Elena Alegret	49.95 %	Scene segmentation
Pascal Fina	49.95 %	Scene geometry
Benedikt Jung	0.1 %	Support with poster
Louis Petrini	0 %	Did not contribute
Chengming Wang	0 %	Withdrew from the course

Abstract

Object-centric 3D representations describe a scene as a set of compact, reusable per-object units, but the pipelines that produce them usually rely on heavy geometric supervision. Object-X [8] is a recent example. It builds expressive per-object embeddings, yet only from ground-truth masks, depth, camera poses and object meshes. Such inputs exist only in annotated RGB-D datasets and are unavailable for in-the-wild RGB sequences, which makes the supervised input stack the main barrier to applying such models classical benchmarks. Our method is an automatic preprocessing pipeline that predicts geometry and segmentation from RGB alone, removing this reliance on ground-truth annotations. It estimates object masks and identities with SAM2-based video segmentation, and predicts camera poses, intrinsics and depth with feed-forward geometry models.

We evaluate two variants, SAM2 with MUST3R and SAM2Object with Pi3X, on 3RScan and ScanNet. RGB-only preprocessing drives the full Object-X pipeline without any

ground-truth annotations, with the decoded objects staying faithful to the predicted geometry. Of the two, SAM2Object with Pi3X performs best overall. The same pipeline also runs in the wild, turning a single handheld RGB video into Object-X-ready inputs.

1. Introduction

Object-centric 3D scene understanding decomposes a scene into reusable object units rather than a single global reconstruction. Such representations benefit editing, retrieval and downstream tasks, but most methods that produce them depend on heavily annotated RGB-D input.

Object-X [8] is a recent instance of this paradigm. It encodes each object into a compact embedding that fuses geometry and appearance and decodes it into renderable 3D Gaussians [5]. The embedding can be further extended to support downstream tasks such as scene alignment and localization. Concretely, Object-X voxelizes each object from its geometry, assigns multi-view DINOv2 features [10] to the voxels, compresses them into a per-object embedding, and decodes that embedding into 3D Gaussians.

Building these inputs requires annotated data. Object-X needs a per-object segmentation to isolate the object, camera poses and intrinsics to project the image features, and per-object geometry to define the voxel grid. Such segmentation, poses and geometry are available only from annotated RGB-D datasets such as 3RScan [14]

and ScanNet [4]. This dependence is the real obstacle to using Object-X outside curated datasets. An ordinary RGB recording provides none of the ground-truth signals it relies on.

We therefore target the input side of Object-X. We ask whether its annotated inputs can be replaced by predictions from RGB alone, while keeping the Object-X model and decoder unchanged.

Approach. We build a preprocessing pipeline that converts an RGB sequence into the per-object inputs Object-X expects. A segmentation branch predicts object masks and identities with SAM2 video segmentation [11]. A geometry branch predicts camera poses, intrinsics and depth with multi-view geometry models (MUST3R [2], Pi3X [17]). We back-project the masked depth into 3D and fuse the lifted points into per-object voxels. From these voxels we assemble the object geometry and scene metadata consumed by Object-X, replacing dataset masks, poses and geometry with predicted inputs. Because it relies only on RGB sequences, the pipeline also runs with captures. We show a qualitative reconstruction from an RGB sequence recorded outside these datasets (Fig. 8).

Contributions.

- We implement an RGB-only preprocessing pipeline that replaces the ground-truth masks, poses and geometry supervised object-centric models require with predicted inputs, instantiated for Object-X.
- We instantiate two variants, SAM2 with MUST3R and SAM2Object with Pi3X, spanning 2D mask propagation and more 3D-consistent object instances.
- We quantify how faithfully decoded Object-X geometry preserves its predicted input, on 3RScan and ScanNet.

Our own work is the end-to-end integration. We convert benchmark data and in-the-wild phone videos into a common scene format and process them with the same RGB-only pipeline to produce the inputs required by Object-X. We implemented the data adapters and connections between modules, masked back-projection, object-level fusion, metadata reconstruction, and the evaluation pipeline.

2. Related Work

Object-centric 3D representations. Neural radiance fields [9] and 3D Gaussian Splatting [5] model a scene as a single field optimised for novel-view synthesis. Object-centric methods instead decompose the scene into per-object units. Object-X [8] encodes each object into a compact, decodable embedding that also serves as a reusable descriptor. Such representations assume curated, object-level inputs typically drawn from annotated RGB-D datasets.

From 2D to 3D-consistent segmentation. Foundation models segment objects in 2D without category supervision. Segment Anything [6] provides promptable image masks, and SAM2 [11] extends this to video, propagating a prompted mask into a track via a memory bank. Video object segmentation methods such as Cutie [3] target the same goal. These predictions are view-inconsistent, so a line of work lifts 2D masks into a multi-view-consistent 3D segmentation. Panoptic Lifting [13] resolves inconsistent instance IDs by linear assignment, Contrastive Lift [1] and OmniSeg3D [21] learn and cluster a 3D instance feature field, Gaussian Grouping [20] associates masks across views to group object-level Gaussians, and SAM2Object [22] consolidates SAM2 masks across frames and clusters superpoints into 3D instances. We build on SAM2Object to recover 3D-consistent object instances from predicted RGB geometry.

Feed-forward geometry from RGB. Classical structure-from-motion such as COLMAP [12] recovers poses and sparse geometry through feature matching and bundle adjustment. Feed-forward models predict geometry directly. DUST3R [16] regresses pointmaps from uncalibrated image pairs, jointly recovering depth and pose. MAST3R [7] adds metric feature matching. MUST3R [2] and VGGT [15] scale the paradigm to many views in a common coordinate frame, predicting poses, intrinsics and depth in a single pass, while Pi3X [17] drops the fixed reference view and predicts affine-invariant poses and local point maps. We decided to use Pi3X and MUST3R to replace the ground-truth poses, intrinsics and depth.

Positioning. Prior work either targets a scene-level radiance field or studies segmentation and geometry prediction in isolation. We instead chain promptable segmentation with feed-forward geometry to produce the structured, object-level inputs an object-centric representation such as Object-X requires, and analyse how errors in these predictions propagate to the final decoded geometry.

3. Method

Fig. 1 gives an overview. From an RGB sequence with no ground-truth masks or geometry, a segmentation branch and a geometry branch predict per-object masks and scene geometry. These are fused into object-level voxels that replace the annotated inputs Object-X normally consumes. We instantiate two variants, SAM2 with MUST3R and SAM2Object with Pi3X.

3.1. Preliminary: Object-X

Object-X is a 3D object representation method that encodes per-object embeddings from images and meshes, and

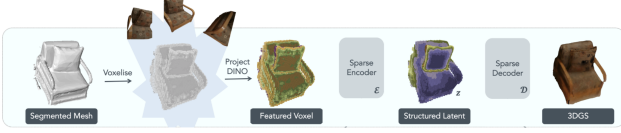


Figure 2. Overview of the Object-X pipeline [8].

decodes them into 3D Gaussian Splatting (3DGS) [5] reconstructions suitable for downstream tasks such as scene alignment and localisation.

Central to Object-X is a structured voxel grid of resolution 64^3 , adapted from TRELLIS [19]. For each object, DINOv2 [10] features extracted from posed RGB frames are back-projected onto the grid using known camera parameters and instance masks resulting in a per-voxel feature aggregation called Featured Voxels. As mentioned, this step requires three supervised inputs: instance segmentation masks, depth and camera poses and intrinsics. Our work targets precisely this bottleneck, predicting all three from RGB video alone.

3.2. Segmentation from RGB

The segmentation branch outputs per-frame object-id masks and an object registry. In the SAM2 with MUST3R variant we treat segmentation as 2D video tracking. We select a small set of diverse keyframes, run the SAM2 automatic mask generator on them, and use the resulting masks as seeds for the SAM2 video predictor, which propagates them across the sequence. Duplicate and fragmented tracks are then merged by spatial overlap.

The SAM2Object with Pi3X variant adds a 3D-consistency stage based on SAM2Object [22]. It clusters superpoints into 3D instances using SAM2 masks and a superpoint affinity graph. SAM2Object normally over-segments a ground-truth mesh, we instead build the superpoints from the predicted Pi3X geometry, run its mask consolidation and graph clustering, and export the resulting 3D instances, with a cleanup step that removes scattered fragments.

3.3. Geometry from RGB

The geometry branch replaces dataset depth, poses and intrinsics with RGB-only predictions. MUST3R [2], run in video mode, provides the global camera trajectory, intrinsics and per-view depth, from which we discard low-confidence pixels. Pi3 [18] predicts higher-resolution camera-frame point maps, but its pose estimates can drift over long indoor sequences; we therefore keep the MUST3R trajectory as a pose prior and use Pi3 only for local surface geometry. Given the MUST3R poses and intrinsics, Pi3 predicts point maps that we scale-align to the pose-prior geometry and filter. We refer to this configuration, MUST3R

poses with Pi3 surface geometry, as *Pi3X*.

3.4. Fusion and Object-X input construction

Object-X builds the per-object voxel grid from the annotated mesh and uses the RGB views only to attach DINOv2 features, by projecting the voxels into each image. As we have no mesh, our fusion stage instead derives the geometry from predicted depth, for each object and view, we back-project the pixels inside the predicted mask with the predicted depth and intrinsics, lift them to world coordinates with the predicted poses, and accumulate them over views. We then remove outliers and weakly supported points, normalise and voxelise into the same 64^3 unit-cube grid as Object-X. Feature attachment follows Object-X unchanged, projecting the voxels into the masked views and averaging the sampled DINOv2 patch features, so the resulting featured voxels form the per-object input its encoder consumes.

4. Experiments

We evaluate how well the pseudo-ground-truth our pipeline predicts can replace the real annotations: the accuracy of the predicted segmentation and geometry against ground truth, and how faithfully Object-X preserves the pseudo-ground-truth instances and the geometry it is given.

Datasets. We evaluate on 3RScan [14] and ScanNet [4]. Both provide RGB-D sequences and reconstructed meshes. Our pipeline uses only the RGB frames. All other modalities (depth, poses, intrinsics, annotations and meshes) serve only as evaluation ground truth. Due to compute constraints, we use a fixed and diverse subset of 100 scenes per dataset and reuse it across all experiments.

Evaluation. We evaluate two variants, SAM2 with MUST3R and SAM2Object with Pi3X, where Pi3X uses MUST3R poses as a prior. We first evaluate the two stages in isolation: the geometry metrics on the MUST3R or Pi3X geometry (Tab. 2), and the segmentation metrics on the ground-truth mesh (Tab. 1). We then evaluate the pipeline end-to-end, where segmentation and geometry are tested together: the pseudo-masks define which objects are present, and the pseudo-geometry gives their shape. We compare the decoded Object-X output against the ground-truth mesh, which measures reconstruction quality, and the voxelized geometry fed into Object-X, which measures how faithfully it preserves its input (Tab. 3).

Geometry evaluation. Predicted depth is lifted into 3D using the predicted poses and intrinsics. Because the resulting point cloud and the dataset mesh may use different coordinate frames, we align them with one

global similarity transform estimated from matching image rays. We then measure bidirectional nearest-neighbour distances. Prediction-to-ground-truth distance measures accuracy, while ground-truth-to-prediction distance measures completeness. We report both directions, their Chamfer-style sum, and precision, recall and F1 at 5 cm.

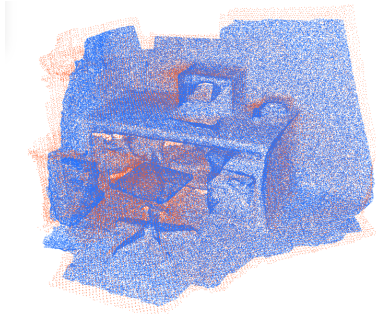


Figure 3. Reconstruction overlaid with ground-truth mesh.

Segmentation evaluation. We evaluate segmentation as a class-agnostic 3D instance-segmentation task. Predicted labels are vertex-aligned with the per-vertex ground-truth instance ids, so overlap is measured directly over mesh vertices. For each scan we compute the IoU between every predicted and every ground-truth instance and match them greedily in decreasing order of IoU. A predicted instance counts as a true positive when its IoU with the matched object is at least a threshold τ . Following the ScanNet protocol, unannotated vertices are ignored, and an unmatched prediction is counted as a false positive only if at most a fraction τ of it falls on these void regions. We pool true positives, false positives and false negatives across all scans and report averaged precision, recall and F1. We report two settings: an *objects-only* setting that excludes the structural classes (wall, floor, ceiling) and an *all-instances* setting that includes them. Tab. 1 reports F1 at $\tau \in \{0.25, 0.5\}$, precision and recall at $\tau = 0.5$.

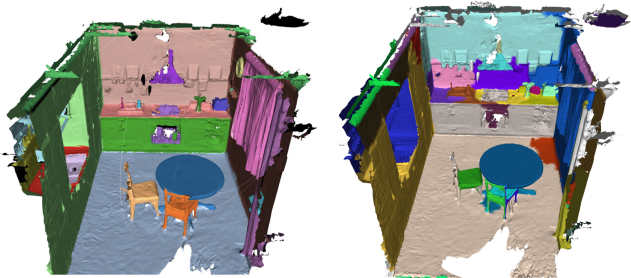


Figure 4. Qualitative results of SAM2Object on 3RScan. Left: input RGB frame with ground-truth instance segmentation. Right: predicted 3D-consistent instance masks from SAM2Object overlaid on the GT geometry.

5. Results

All geometry numbers are robust medians over the scenes each method completed, and arrows in the tables indicate whether higher ($\uparrow$) or lower ($\downarrow$) is better. Each metric is defined in its table caption.

Segmentation. Segmentation is markedly stronger on ScanNet than on 3RScan (Tab. 1), and excluding structural surfaces improves both. The objects-only gain is expected: walls, floors and ceilings are large and hard to split into clean instances, so removing them isolates how well true objects are separated. The large $F1@0.25 \rightarrow F1@0.5$ drop shows that instances are mostly found but only loosely localised: the predicted masks roughly cover each object, yet their boundaries are not tight enough for the stricter overlap. On 3RScan recall is the main bottleneck, with many objects missed altogether, likely reflecting its noisier, lower-quality reconstructions; ScanNet, by contrast, keeps precision and recall balanced.

Dataset	Subset	F1@.25 $\uparrow$	F1@.50 $\uparrow$	Prec.@.50 $\uparrow$	Rec.@.50 $\uparrow$
3RScan	all	0.40	0.21	0.28	0.17
3RScan	objects-only	0.49	0.30	0.44	0.23
ScanNet	all	0.57	0.41	0.36	0.48
ScanNet	objects-only	0.67	0.52	0.49	0.55

Table 1. Class-agnostic 3D instance segmentation on the ground-truth meshes. F1, precision and recall are reported at IoU thresholds of 0.25 and 0.50; the *objects-only* subset excludes the structural classes (wall, floor, ceiling) and *all* includes them.

Geometry. On 3RScan the two methods are essentially tied, while Pi3X comes out ahead on ScanNet, and both do better on ScanNet than on 3RScan (Tab. 2). As with segmentation, the dataset matters more than the method. Recall stays high, so the geometry covers most of the surface, but precision is the limiting factor: some predicted points end up off the true surface. Pi3X only gains where the input is cleaner, which suggests its higher-resolution geometry helps mainly when the scene is already reliable.

Dataset	Method	Dist.(Pred,GT) $\downarrow$	Dist.(GT, Pred) $\downarrow$	Chamfer $\downarrow$
3RScan	MUS3R	0.0525	0.0197	0.0722
3RScan	Pi3X	0.0537	0.0204	0.0741
ScanNet	MUS3R	0.0470	0.0164	0.0634
ScanNet	Pi3X	0.0419	0.0162	0.0580

Dataset	Method	P@5cm $\uparrow$	R@5cm $\uparrow$	F1@5cm $\uparrow$
3RScan	MUS3R	0.482	0.787	0.598
3RScan	Pi3X	0.472	0.764	0.584
ScanNet	MUS3R	0.519	0.914	0.662
ScanNet	Pi3X	0.553	0.893	0.683

Table 2. Input geometry quality before running the end-to-end pipeline. Dist.(Pred,GT) and Dist.(GT, Pred) are in meters.

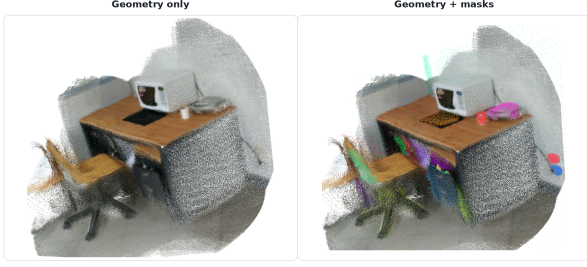


Figure 5. Reconstruction of SAM2 & MUST3R



Figure 6. Reconstruction of SAM2Object & Pi3X

End-to-end pipeline. Tab. 3 evaluates the full pipeline, where the predicted segmentation and geometry are passed through Object-X together. Per variant we report *Final vs. GT* (decoded output against the dataset mesh) and *Final vs. Object-X input* (decoded output against the voxelized input), capturing the joint effect of predicted masks and geometry.

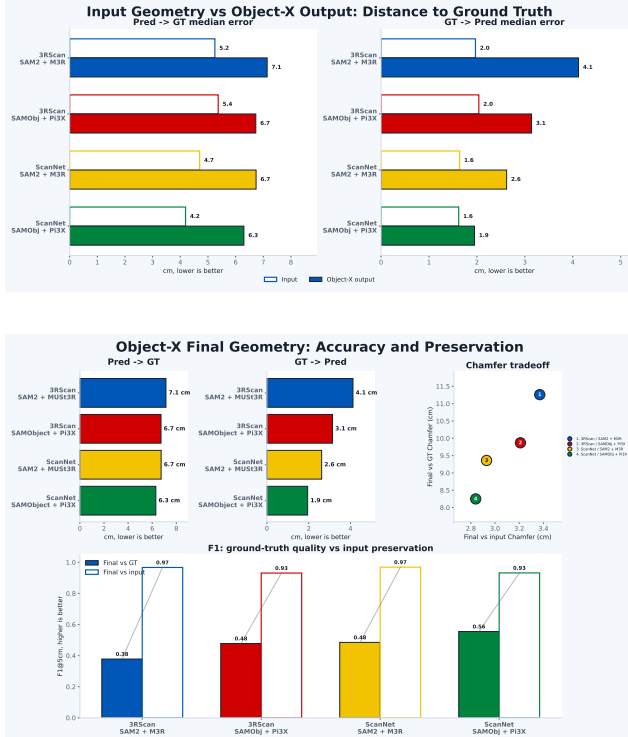


Figure 7. Object-X geometry. Top: input-to-output preservation. Bottom: directional errors to ground truth.

Fig. 7 makes the trend explicit. In the paired directional-error plot, the outlined bars show the predicted input geometry measured against ground truth, while the filled bars show the final Object-X output against the same ground truth. The output error is higher for all four configurations, showing that some accuracy and surface coverage are lost as the geometry passes through segmentation, object fusion, voxelization and decoding. This loss is most visible in the ground-truth-to-prediction direction, especially for SAM2 with MUST3R on 3RScan.

The summary plot separates this loss from input preservation. Although F1 against ground truth is lower, F1 between the final output and the geometry given to Object-X remains high. The decoder therefore retains most of its voxelized input rather than replacing it with unrelated geometry. The Chamfer plot combines both aspects, where points closer to the lower-left corner have lower ground-truth error and better input preservation. ScanNet with SAM2Object and Pi3X gives the strongest overall result. Both ScanNet variants lie closer to the lower-left corner than their 3RScan counterparts, and within each dataset SAM2Object with Pi3X performs best. SAM2 with MUST3R remains competitive in prediction-to-ground-truth distance but loses more reference-surface coverage, as reflected by its lower recall.

Finally, we also tested the pipeline on our own handheld room capture (Fig. 8), for which no dataset annotations, meshes or camera poses are available. This qualitative example shows the intended end-to-end use case. From an ordinary RGB video, the system recovers a coherent scene geometry and predicted object masks.

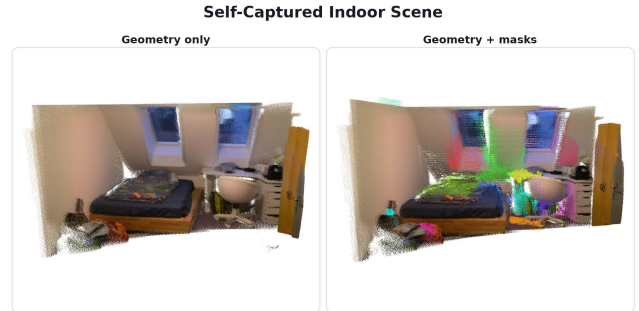


Figure 8. End-to-end result on our own handheld room capture. Left: reconstructed geometry only. Right: the same reconstruction with predicted object masks.

	Variant	Dist.(Pred,GT) ↓	Dist.(GT,Pred) ↓	P@5cm ↑	R@5cm ↑	F1@5cm ↑
<i>Output vs. GT mesh</i>						
3RScan	SAM2 / MUST3R	0.0714	0.0412	0.371	0.387	0.378
	SAM2Object / Pi3X	0.0673	0.0314	0.393	0.611	0.478
ScanNet	SAM2 / MUST3R	0.0674	0.0262	0.392	0.636	0.485
	SAM2Object / Pi3X	0.0630	0.0195	0.416	0.835	0.555
<i>Output vs. Object-X input</i>						
3RScan	SAM2 / MUST3R	0.0202	0.0135	0.992	0.943	0.967
	SAM2Object / Pi3X	0.0194	0.0127	0.898	0.968	0.931
ScanNet	SAM2 / MUST3R	0.0183	0.0109	0.976	0.962	0.969
	SAM2Object / Pi3X	0.0165	0.0119	0.891	0.976	0.932

Table 3. Final Object-X geometry results. Distances are in metres. Bold indicates the best variant per dataset within each comparison block.

6. Conclusion

What holds Object-X back in practice is not its object-centric representation but the supervised input it expects: posed and calibrated views, object masks and per-object mesh geometry. We remove that requirement with an RGB-only preprocessing pipeline that predicts masks, object identities, camera poses and intrinsics, depth and object geometry, and assembles them into the scene metadata Object-X consumes. On 3RScan and ScanNet the pipeline runs end to end without any ground-truth input, and the decoded reconstructions stay close to the geometry we feed in. Importantly, this is not limited to benchmark datasets such as 3RScan or ScanNet. We also recorded our own room video with a phone and ran the same pipeline on it. The resulting reconstruction with object masks (Fig. 8) underlines that Object-X no longer needs dataset-provided ground-truth inputs, but can be driven from an ordinary RGB video.

SAM2Object with Pi3X is the strongest variant overall, while SAM2 with MUST3R stays competitive but loses more surface coverage. The pattern points to where effort is best spent. Not on the Object-X decoder, which largely preserves what it is given, but on the predicted inputs. More 3D-consistent segmentation, geometry-aware mask filtering and refinement of poses and depth would feed through directly to the final objects. With those, Object-X stops depending on dataset-provided metadata and becomes a usable, annotation-free route to object-centric 3D reconstruction from RGB alone.

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